

# Piston Perfection

By the mid-1940s, it was clear that jet engines were going to revolutionize aircraft design and take over in many arenas, but piston engines continued to be refined and to excel in specific fields. With lower fuel consumption than early jets, they remained ideal for ultra-long-range aircraft; they were also better suited to naval carrier operations, and for seaplanes.



## ◁ Supermarine Seafire F MkXVII 1941

**Origin** UK

**Engine** 1,850-2,375 hp Rolls-Royce Merlin/Griffon supercharged liquid-cooled V12

**Top speed** 387 mph (623 km/h)

Conceived in 1939, the seaborne Spitfire was delayed at first. Progressively improved with rocket-assisted takeoff, folding wings, and ever more power, it remained fragile for carrier use.



## △ Yakovlev Yak-9 1942

**Origin** USSR

**Engine** 1,650 hp Klimov VK-107A supercharged liquid-cooled V12

**Top speed** 435 mph (700 km/h)

Light, fast, easy to fly, and progressively updated, this was the most-produced Soviet fighter with 16,769 built. Its small, high-speed wing required long takeoff and landing runs.



## △ Northrop P-61 Black Widow 1943

**Origin** US

**Engine** 2 x 2,250 hp Pratt & Whitney R-2800-5W Double Wasp air-cooled 18-cylinder radial

**Top speed** 366 mph (589 km/h)

The first night-interceptor built by the US, and the first aircraft specifically designed to carry radar, the Black Widow could stay aloft for up to eight hours. It operated extensively in WWII.



## △ Grumman F7F-3 Tigercat 1944

**Origin** US

**Engine** 2 x 2,100 hp Pratt & Whitney R-2800-34W Double Wasp air-cooled 18-cylinder radial

**Top speed** 460 mph (740 km/h)

The US Navy's first twin-engine fighter entered service at the end of WWII. One of the fastest piston-engined fighters, it served widely in Korea, both in land-based and carrier operations.

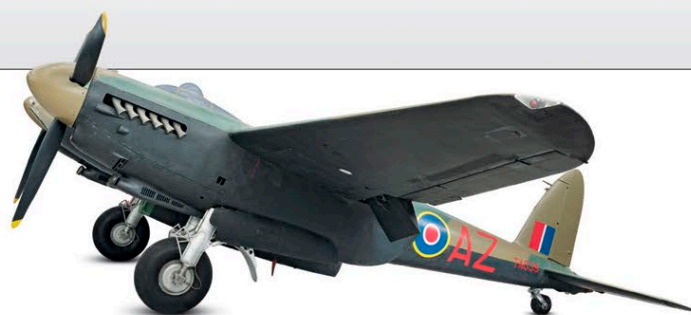
## ◁ Grumman F8F Bearcat 1944

**Origin** US

**Engine** 2,100 hp (later 2,250 hp) Pratt & Whitney R-2800 "Double Wasp" air-cooled 18-cylinder radial

**Top speed** 421 mph (678 km/h)

The last of the piston-engined "Cats," the F8F was 20 percent lighter than the F6F Hellcat, climbed 30 percent quicker, and was 40 mph (64 km/h) faster. It served in Korea and holds piston-engined speed records.



## △ de Havilland DH98 Mosquito 1940

**Origin** UK

**Engine** 2 x 1,480 hp Rolls-Royce Merlin 21/21 + 23/23 water-cooled V12 (later 2 x 1,690 hp 113 + 114)

**Top speed** 366-415 mph (589-670 km/h)

Built entirely of wood, the "Wooden Wonder" was the world's fastest military aircraft in 1941. Conceived as an unarmed fast bomber, it fulfilled many roles from photo reconnaissance to fighter.



## △ Lavochkin La-5 1942

**Origin** USSR

**Engine** 1,850 hp ShvetsovASH-82FN air-cooled 14-cylinder radial

**Top speed** 403 mph (650 km/h)

This effective Russian WWII combat aircraft, built-like the Mosquito—in wood, became a match for German fighters at low altitude once fuel injection was added; 9,920 were built.

